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Lecture 1: Optimal Transport

Lecture 2: Generative Flows

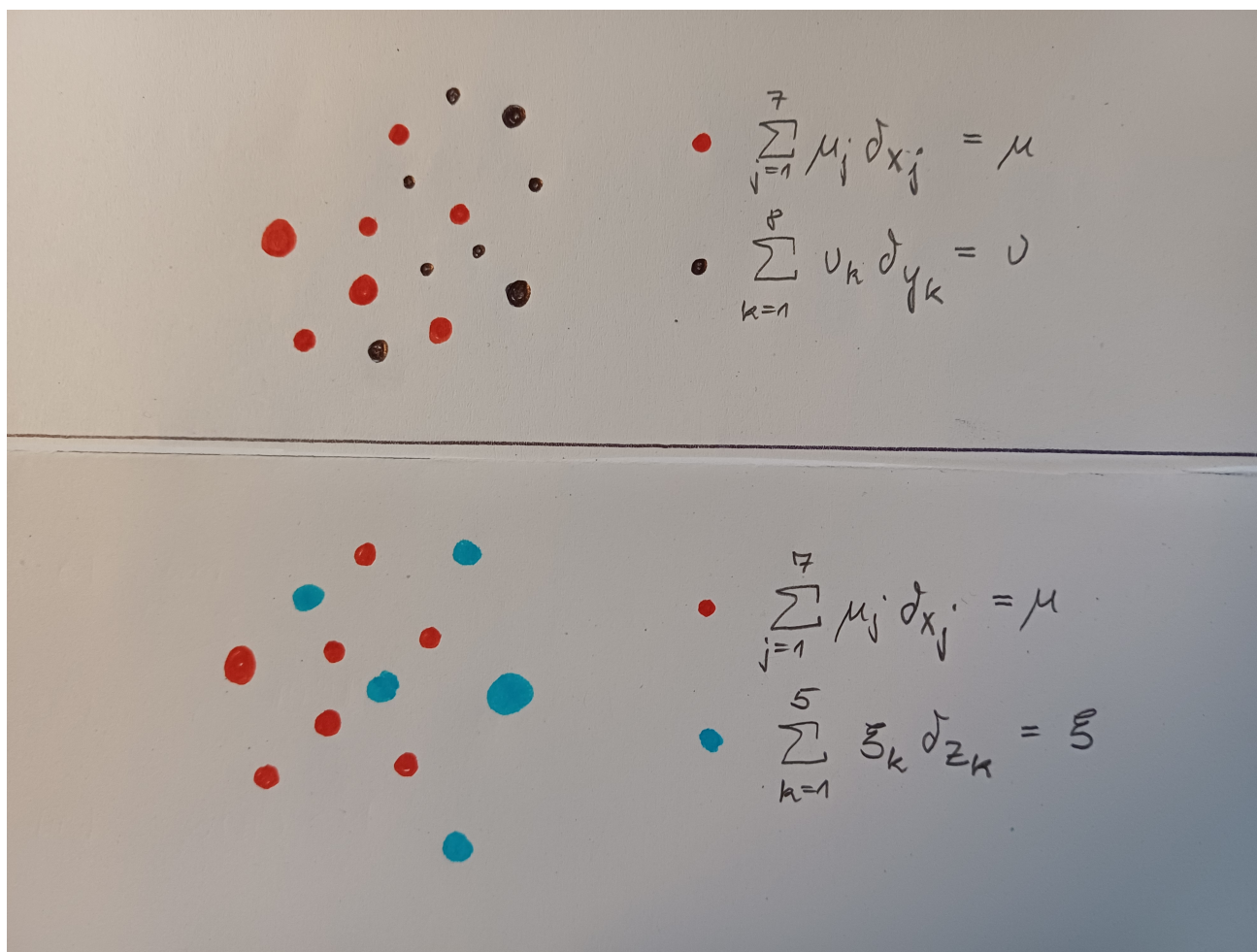
Lecture 3: From 1D Processes to Flows

Lecture 4: Bayesian Inverse Problems

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Question:

- ◆ What is the best way to transport the mass from μ to ν and what costs such a transport?
- ◆ Is ν or ξ nearer to μ ?



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1. Optimal Transport (OT)

1. Spaces of Measures
2. Monge and Kantorovich Problem of OT
3. Discrete OT and its Dual
4. Continuous OT
5. Regularized Optimal Transport

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1. Spaces of Measures

- ◆ $\mathcal{M}(\mathbb{R}^d)$ is a Banach space with **total variation norm**

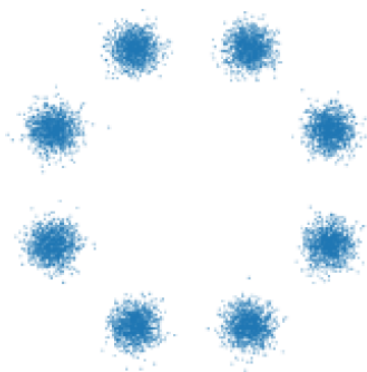
$$\|\mu\|_{TV} := |\mu|(\mathbb{R}^d) \quad \text{where} \quad |\mu|(A) := \sup_{A = \bigcup_{k=1}^n A_k, A_i \cap A_j = \emptyset} \sum_{k=1}^n |\mu(A_k)|$$

- ◆ pre-dual space of $\mathcal{M}(\mathbb{R}^d)$ is $C_0(\mathbb{R}^d)$, i.e. $C_0(\mathbb{R}^d)' = \mathcal{M}(\mathbb{R}^d)$

$$\|\mu\|_{TV} = \sup_{\|\varphi\|_\infty \leq 1} |\langle \varphi, \mu \rangle|, \quad \langle \varphi, \mu \rangle := \int_{\mathbb{R}^d} \varphi(x) d\mu(x)$$

- ◆ $\mathcal{P}(\mathbb{R}^d)$ subset of probability measures on $\mathbb{R}^d$

Samples from a probability measure:



$d = 2$



$d = 28^2 = 784$

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Examples: $\mathcal{P}(\mathbb{R}^d)$

1. Atomic measures (empirical measures if same weights)

$$\mu = \sum_{i=1}^m \mu_i \delta_{x_i}, \quad \delta_x(A) := \begin{cases} 1 & \text{if } x \in A, \\ 0 & \text{otherwise} \end{cases}$$

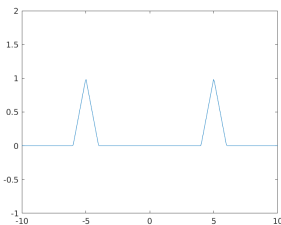
$$\|\mu\|_{TV} = |\mu_1| + \dots + |\mu_m| = 1$$

2. Absolutely continuous measures $\mu \in \mathcal{M}(\mathbb{R}^d)$ with density $\varphi \in L_1(\mathbb{R}^d)$

$$\mu(A) = \int_A \varphi(x) \, dx$$

$$\|\mu\|_{TV} = \int_{\mathbb{R}^d} |\varphi(x)| \, dx = 1$$

◆ Also, if μ and ν have disjoint supports $\|\mu - \nu\|_{TV} = 2$



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2. Monge Problem (1781)



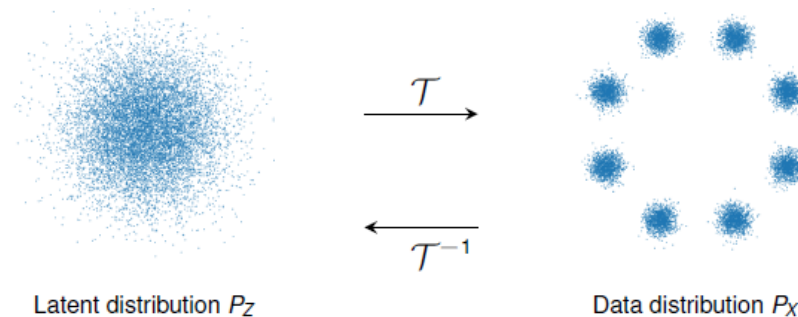
Given $\mu \in \mathcal{P}(\mathbb{R}^d)$, $\nu \in \mathcal{P}(\mathbb{R}^d)$

Find an optimal **transport map** $\hat{T} : \mathbb{R}^d \rightarrow \mathbb{R}^d$ such that

$$\hat{T} \in \operatorname{argmin}_{T \text{ measurable}} \int_X c(x, T(x)) \, d\mu(x) \quad \text{subject to} \quad \nu = T_{\#}\mu$$

with the **push forward measure**

$$T_{\#}\mu := \mu \circ T^{-1}$$



$$\diamond \int_{T^{-1}(A)} f(T(x)) d\mu(x) = \int_A f(y) d \underbrace{(T_{\#}\mu)}_{\nu}(y)$$

◆ in case of density $\mu = p_{\mu} d\lambda$ and a diffeomorphism T :

$$p_{T_{\#}\mu}(y) = p_{\mu}(T^{-1}(y)) |\det \nabla T^{-1}(y)|$$

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Discrete Monge Problem

Given

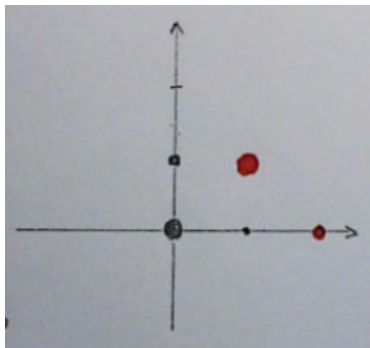
$$\mu = \sum_{i=1}^m \mu_i \delta_{x_i}, \quad \nu = \sum_{j=1}^n \nu_j \delta_{y_j}$$

Find an optimal transport map $\hat{T} : \{x_1, \dots, x_m\} \rightarrow \{y_1, \dots, y_n\}$ such that

$$\hat{T} \in \operatorname{argmin}_T \sum_{i=1}^N c(x_i, T(x_i)) \mu_i \quad \text{s.t.} \quad \nu_j = \sum_{T(x_i)=y_j} \mu_i$$

Example: $x_1 = (0, 0)$, $x_2 = (1, 0)$, $x_3 = (0, 1)$ and $y_1 = (1, 1)$, $y_2 = (2, 0)$

Measures: $\mu = 3\delta_{x_1} + 1\delta_{x_2} + 2\delta_{x_3}$, $\nu = 4\delta_{y_1} + 2\delta_{y_2}$



Costs: $c(x, y) = \|x - y\|^2$

Only possible map: $T(x_1) = y_1$, $T(x_2) = y_1$, $T(x_3) = y_2$

Problem: Opposite task, i.e. changing the role of μ and ν , has no solution T

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Example: For two Gaussians $\mu = \mathcal{N}(m_\mu, \Sigma_\mu)$ and $\nu = \mathcal{N}(m_\nu, \Sigma_\nu)$, where $\mathcal{N}(m, \Sigma)$ has the density

$$p(x) := (2\pi)^{-\frac{d}{2}} (\det \Sigma)^{-\frac{1}{2}} e^{-\frac{1}{2}(x-m)^\top \Sigma^{-1}(x-m)},$$

the transport map is given by

$$T(x) = m_\nu + A(x - m_\mu), \quad A := \Sigma_\mu^{-\frac{1}{2}} \left(\Sigma_\mu^{\frac{1}{2}} \Sigma_\nu \Sigma_\mu^{\frac{1}{2}} \right) \Sigma_\mu^{-\frac{1}{2}},$$

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Kantorovich Problem (1942)



Given $\mu \in \mathcal{P}(\mathbb{R}^d)$, $\nu \in \mathcal{P}(\mathbb{R}^d)$

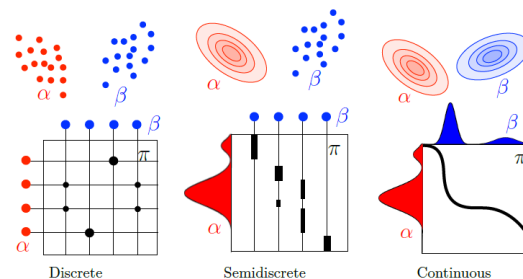
Find an optimal **transport plan** $\hat{\alpha} \in \mathcal{P}(\mathbb{R}^d \times \mathbb{R}^d)$ with given marginals μ and ν

$$\hat{\pi} \in \operatorname{argmin}_{\pi \in \Pi(\mu, \nu)} \int_{\mathbb{R}^d \times \mathbb{R}^d} c(x, y) d\alpha(x, y)$$

$$\operatorname{OT}(\mu, \nu) = \min_{\alpha \in \Pi(\mu, \nu)} \int_{\mathbb{R}^d \times \mathbb{R}^d} c(x, y) d\alpha(x, y)$$

where

- ◆ $\Pi(\mu, \nu) := \{\alpha \in \mathcal{P}(\mathbb{R}^d \times \mathbb{R}^d) : (P_1)_\# \alpha = \mu, (P_2)_\# \alpha = \nu\}$
- ◆ $P_1(x_1, x_2) = x_1, P_2(x_1, x_2) = x_2$



: Book Peyré/Cuturi: Computational optimal transport

- ◆ **Existence** of a minimizer is ensured if c is lsc and bounded from below

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Given

$$\mu = \sum_{i=1}^m \mu_i \delta_{x_i}, \quad \nu = \sum_{j=1}^n \nu_j \delta_{y_j}$$

Find an optimal **transport plan** $\hat{\alpha} = \sum_{i,j=1}^{m,n} \hat{\alpha}_{ij} \delta_{x_i, y_j}$, i.e. $\alpha = (\alpha_{ij})_{i,j} \in \mathbb{R}_{\geq 0}^{m,n}$ such that

$$\hat{\alpha} \in \operatorname{argmin}_{\alpha} \sum_{i=1}^m \sum_{j=1}^n c(x_i, y_j) \alpha_{ij} = \operatorname{argmin}_{\alpha} \langle c, \alpha \rangle$$

subject to $\alpha_{ij} \geq 0$ and

$$\sum_{i=1}^m \alpha_{ij} = \nu_j, \quad j = 1, \dots, n, \quad \alpha^T \mathbf{1}_m = \nu$$

$$\sum_{j=1}^n \alpha_{ij} = \mu_i, \quad i = 1, \dots, m, \quad \alpha \mathbf{1}_n = \mu$$

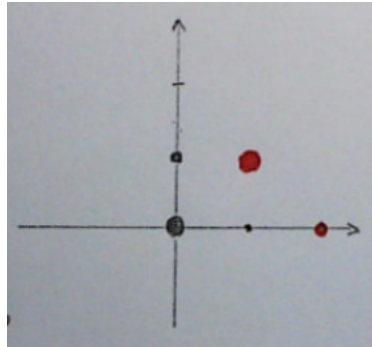
	ν_1	$\dots$	ν_j	$\dots$	ν_m
μ_1	$\alpha_{1,1}$	$\dots$	$\alpha_{1,j}$	$\dots$	$\alpha_{1,m}$
$\vdots$			$\vdots$		$\vdots$
μ_i	$\alpha_{i,1}$	$\dots$	$\alpha_{i,j}$	$\dots$	$\alpha_{i,m}$
$\vdots$			$\vdots$		$\vdots$
μ_n	$\alpha_{n,1}$	$\dots$	$\alpha_{n,j}$	$\dots$	$\alpha_{n,m}$

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Examples with $c(x, y) = \|x - y\|^2$

Example 1: $x_1 = (0, 0)$, $x_2 = (1, 0)$, $x_3 = (0, 1)$ and $y_1 = (1, 1)$, $y_2 = (2, 0)$

Measures: $\mu = 3\delta_{x_1} + 1\delta_{x_2} + 2\delta_{x_3}$, $\nu = 4\delta_{y_1} + 2\delta_{y_2}$



Optimal map: $T(x_1) = y_1$, $T(x_2) = y_1$, $T(x_3) = y_2$

Optimal plan:

	y_1	y_2
x_1	2	4
x_2	1	1
x_3	1	5

costs

	4	2
3	2	1
1	0	1
2	2	0

optimal plan $\hat{\alpha}$

$$\text{OT}(\mu, \nu) = 11$$

	4	2
3	3	0
1	1	0
2	0	2

$$\alpha = (\text{Id}, T)_{\#}\mu$$

$$\langle c, \alpha \rangle = 17$$

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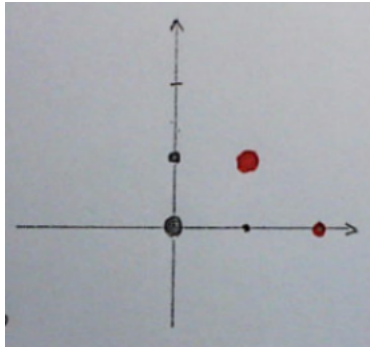
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Examples with $c(x, y) = \|x - y\|^2$

Example 2: $x_1 = (0, 0)$, $x_2 = (1, 0)$, $x_3 = (0, 1)$ and $y_1 = (1, 1)$, $y_2 = (2, 0)$

Measures: $\mu = 3\delta_{x_1} + 1\delta_{x_2} + 2\delta_{x_3}$, $\nu = 2\delta_{y_1} + 4\delta_{y_2}$



Optimal map: $T(x_1) = y_2$, $T(x_2) = y_2$, $T(x_3) = y_1$

Optimal plan:

	y_1	y_2
x_1	4	2
x_2	1	1
x_3	5	1

costs

	2	4
3	0	3
1	0	1
2	2	0

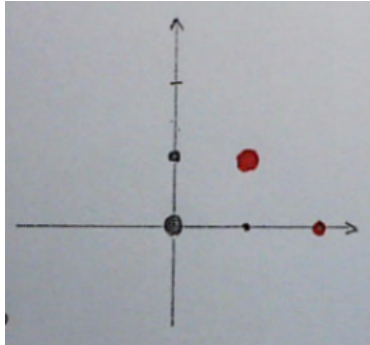
optimal plan $\hat{\alpha} = (\text{Id}, T)_{\#}\mu$

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Examples with $c(x, y) = \|x - y\|^2$

Example 3: $x_1 = (1, 1)$, $x_2 = (2, 0)$ and $y_1 = (0, 0)$, $y_2 = (1, 0)$, $y_3 = (0, 1)$ and $y_1 = (1, 1)$, $y_2 = (2, 0)$

Measures: $\mu = 2\delta_{x_1} + 4\delta_{x_2}$, $\nu = 3\delta_{y_1} + 1\delta_{y_2} + 2\delta_{y_3}$



Optimal map: does not exist

Optimal plan:

	y_1	y_2	y_3
x_1	4	1	5
x_2	2	1	1

costs

	3	1	2
2	0	0	2
4	3	1	2

optimal plan $\hat{\alpha}$

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For $p \in [1, \infty)$, the space of measures with finite p -th moment is defined by

$$\mathcal{P}_p(\mathbb{R}^d) := \left\{ \mu \in \mathcal{P}(\mathbb{R}^d) : \int_{\mathbb{R}^d} \|x\|^p d\mu(x) < \infty \right\}.$$

For $\mu, \nu \in \mathcal{P}_p(\mathbb{R}^d)$, the **Wasserstein p -distance** is given by

$$W_p^p(\mu, \nu) := \inf_{\pi \in \Pi(\mu, \nu)} \int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|^p d\alpha(x, y),$$

which actually defines a **metric**.

The metric space $(\mathcal{P}_p(\mathcal{X}), W_p)$ is called the p -th **Wasserstein space**.

Convergence of $(\mu_n)_n$, $\mu_n \in \mathcal{P}_2(\mathbb{R}^d)$: $W_2(\mu_n, \mu) \rightarrow 0$ as $n \rightarrow \infty$ if and only if we have weak/narrow convergence

$$\mu_n \xrightarrow{*} \mu \text{ in } C_b(X)'$$

and

$$\int_{\mathbb{R}^d} \|x\|^2 d\mu_n(x) \rightarrow \int_{\mathbb{R}^d} \|x\|^2 d\mu(x) \quad \text{as } n \rightarrow \infty$$

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Theorem Let $\mu \in \mathcal{P}_2(\mathbb{R}^d)$, where $\mu \ll \lambda$ and $c(x, y) = \|x - y\|^2$. Then

- ◆ Kantorovich problem has a unique solution $\hat{\pi}$
- ◆ $\hat{\pi} = (I, \hat{T})_{\#}\mu$, where $\hat{T} \in L^2_{\mu}(X, X)$ is the optimal transport map
- ◆ If ν has bounded support, then

$$\hat{T}(x) = x - \nabla \varphi(x) = \nabla \psi(x) \quad \text{for } \mu\text{-a.e. } x,$$

for some lower semi-continuous, convex, differentiable μ -a.e function ψ .

- ◆ Conversely, if ψ is lower semi-continuous, convex and differentiable μ -a.e. with $|\nabla \psi| \in L^2_{\mu}(X, X)$, then

$$T = \nabla \psi$$

is the optimal transport map from μ to $\nu = T_{\#}\mu \in \mathcal{P}_2(X)$.

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1. Basic Notation: Spaces of Measures
2. Monge and Kantorovich Problem of OT
3. Dual OT
4. Regularized Optimal Transport

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Discrete OT and its Dual

(P) $\operatorname{argmin}_{\alpha} \langle c, \alpha \rangle$ subject to $\alpha 1_n = \mu, \alpha^\top 1_m = \nu, \alpha \geq 0$ nm variables

(P) $\operatorname{argmin}_{\alpha} \langle c, \alpha \rangle$ subject to $\underbrace{(1_n \otimes I_m)}_{P_1} \alpha = \mu, \underbrace{(I_n \otimes 1_m)}_{P_2} \alpha = \nu, \alpha \geq 0$

$$P_1 = (I_m \quad \dots \quad I_m) \in \mathbb{R}^{m \times nm}$$

$$P_2 = \begin{pmatrix} \underbrace{1 \dots 1}_m & 0 & \dots & 0 \\ 0 & \underbrace{1 \dots 1}_m & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \underbrace{1 \dots 1}_m \end{pmatrix} \in \mathbb{R}^{n \times nm},$$

(P) $\min_{\alpha \geq 0} \max_{\varphi, \psi} \langle c, \alpha \rangle + \langle \mu - P_1 \alpha, \varphi \rangle + \langle \nu - P_2 \alpha, \psi \rangle = \min_{\alpha \geq 0} \max_{\varphi, \psi} L(\alpha, \mu, \nu)$

(D) $\max_{\varphi, \psi} \min_{\alpha \geq 0} \langle c, \alpha \rangle + \langle \mu - P_1 \alpha, \varphi \rangle + \langle \nu - P_2 \alpha, \psi \rangle$

$$= \max_{\varphi, \psi} \min_{\alpha \geq 0} \langle c - P_1^\top \varphi - P_2^\top \psi, \alpha \rangle + \langle \mu, \varphi \rangle + \langle \nu, \psi \rangle =$$

$$= \max_{\varphi_i + \psi_j \leq c_{i,j}} \langle \mu, \varphi \rangle + \langle \nu, \psi \rangle \quad \text{color: red } n + m \text{ variables}$$

If $(c - P_1^\top \varphi - P_2^\top \psi)_{ij} = c_{ij} - \varphi_i - \psi_j < 0$, then first part = $-\infty$

If $(c - P_1^\top \varphi - P_2^\top \psi)_{ij} \geq 0$, then first = 0, where $\alpha_{ij} = 0$ if $c_{ij} > \varphi_i - \psi_j$

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OT and its Dual (Continuous Setting)

Primal OT:

$$\text{OT}(\mu, \nu) := \min_{\pi \in \Pi(\mu, \nu)} \int_{\mathbb{R}^d \times \mathbb{R}^d} c \, d\pi,$$

Dual OT:

$$\text{OT}(\mu, \nu) = \max_{\substack{(\varphi, \psi) \in C(\mathbb{R}^d)^2 \\ \varphi(x) + \psi(y) \leq c(x, y)}} \int_{\mathbb{R}^d} \varphi \, d\mu + \int_{\mathbb{R}^d} \psi \, d\nu.$$

$(\hat{\varphi}, \hat{\psi}) = (\hat{\varphi}, \hat{\varphi}^c)$ with c -transformed function

$$\varphi^c(y) = \min_{x \in \mathbb{R}^d} \{c(x, y) - \varphi(x)\}.$$

Special case: If $c(x, y) = |x - y|$, then $\varphi = -\psi$ is 1-Lipschitz (more general $c = d$ for metric space with distance d)

$$W_1(\mu, \nu) = \max_{\varphi \in \text{Lip}_1(\mathbb{R}^d)} \int_{\mathbb{R}^d} \varphi \, d\mu = \max_{\varphi \in \text{Lip}_1(\mathbb{R}^d)} \langle \varphi, \mu \rangle$$

This plays a role in so-called [Wasserstein GANs](#).

Compare to the dual notation of the TV norm:

$$\|\mu\|_{TV} = \sup_{\|\varphi\|_{\infty} \leq 1} |\langle \varphi, \mu \rangle|$$

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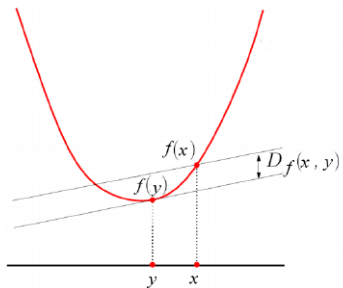
Numerical computation of OT is time consuming for many samples $m = n$:

- Simplex algorithm $O(n^3 \log n)$
- Hungarian algorithm $O(n^3)$
- regularized Wasserstein $O(n^2 \log n)$, critical for small regularization parameter ε
- Toolboxes:
 - Python Optimal Transport (POT, <https://pythonot.github.io/>): general toolbox for optimal transport including unbalanced optimal transport, sliced Wasserstein und Gromov-Wasserstein
 - Geomloss (<https://www.kernel-operations.io/geomloss/>): entropic optimal transport with efficient GPU implementation and automatic differentiation

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Bregman distance: $f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ convex lsc and $\text{dom} f \cap (0, +\infty) \neq \emptyset$

$$D_f(x, y) = f(x) - f(y) + \langle \nabla f(y), x - y \rangle$$



Function Name	$f(x)$	$\text{dom } f$	$D_f(x, y)$
Squared Norm	$\frac{1}{2}x^2$	$(-\infty, \infty)$	$\frac{1}{2}(x - y)^2$
Shannon Entropy	$x \log x$	$[0, \infty)$	$x \log \frac{x}{y} - x + y$
Bit Entropy	$x \log x + (1 - x) \log(1 - x)$	$[0, 1]$	$x \log \frac{x}{y} + (1 - x) \log \frac{1-x}{1-y}$
Burg Entropy	$-\log x$	$(0, \infty)$	$\frac{x}{y} - \log \frac{x}{y} - 1$

Examples:

1. $f(x) = \frac{1}{2}\|x\|^2$

$$D_f(x, y) = \frac{1}{2}\|x\|^2 - \frac{1}{2}\|y\|^2 - \langle y, x - y \rangle = \frac{1}{2}\|x - y\|^2$$

2. $f(x) = \langle x, \log x \rangle$ (negative) Shannon entropy = maximal chaos

$$\begin{aligned} D_f(x, y) &= \langle x, \log x \rangle - \langle y, \log y \rangle - \langle \log y + 1, x - y \rangle \\ &= \langle x, \log x \rangle - \langle x, \log y \rangle - \langle x, 1 \rangle + \langle y, 1 \rangle \\ &= \text{KL}(x, y) \end{aligned}$$

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Properties:

- ◆ $D_f(x, y) \geq 0$
- ◆ $D_f(x, y) = 0$ iff $x = y$ in case f is strictly convex.
- ◆ In general not symmetric and does not fulfill a triangular inequality
- ◆ Jointly convex, lsc.
- ◆ If f is strictly convex, D_f is strictly convex in the first argument.
- ◆ If expressions exist

$$\nabla_x D_f(x, y) = \nabla f(x) - \nabla f(y)$$

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$$(P) \quad OT_\varepsilon = \min_{\alpha \in \Pi(\mu, \nu)} \int_{\mathbb{R}^d \times \mathbb{R}^d} c(x, y) d\alpha(x, y) + \varepsilon \text{KL}(\alpha, \mu \otimes \nu)$$

Alternatively, we can regularize with the entropy

$$\text{ent}(\alpha) = \int_{\mathbb{R}^d \times \mathbb{R}^d} \alpha \log(\alpha) dx(x, y)$$

Discrete setting:

$$\begin{aligned} (P) \quad OT_\varepsilon(\mu, \nu) &= \min_{\alpha \in \Pi(\mu, \nu)} \langle c, \alpha \rangle + \varepsilon \left(\sum_{i,j} \alpha_{i,j} \log \alpha_{i,j} - \alpha_{i,j} \log(\mu_i \nu_j) - \alpha_{i,j} \right) \\ &= \varepsilon \min_{\alpha \in \Pi(\mu, \nu)} \sum_{i,j} \alpha_{i,j} \log \alpha_{i,j} - \alpha_{i,j} \log(\mu_i \nu_j e^{-c/\varepsilon}) - \alpha_{i,j} \\ &= \varepsilon \min_{\alpha \in \Pi(\mu, \nu)} \text{KL}(\alpha, \underbrace{\text{diag}(\mu) e^{-c/\varepsilon} \text{diag}(\nu)}_K) \end{aligned}$$

Rewriting this into the dual form leads to

$$(P) \quad \min_{\alpha} \max_{\varphi, \psi} \text{KL}(\alpha, K) + \langle \mu - P_1 \alpha, \varphi \rangle + \langle \nu - P_2 \alpha, \psi \rangle$$

$$\begin{aligned} (D) \quad & \max_{\varphi, \psi} \min_{\alpha} \{ \text{KL}(\alpha, K) - \langle \alpha, P_1^\top \varphi + P_2^\top \psi \rangle \} + \langle \mu, \varphi \rangle + \langle \nu, \psi \rangle \\ &= \max_{\varphi, \psi} \langle K, e^{-P_1^\top \varphi - P_2^\top \psi} \rangle + \langle \mu, \varphi \rangle + \langle \nu, \psi \rangle \end{aligned}$$

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Sinkhorn Algorithm (Primal)

$$(P) \quad \text{OT}_\varepsilon(\mu, \nu) = \varepsilon \min_{\alpha \in \Pi(\mu, \nu)} \text{KL}(\alpha, K)$$

Sinkhorn algorithm wrt α :

$$\alpha^{(0)} := K,$$

for $r = 0, 1, \dots$

$$\alpha^{(r+\frac{1}{2})} := \text{diag} \left(\frac{\mu}{\alpha^{(r)} \mathbf{1}} \right) \alpha^{(2r)}$$

$$\alpha^{(r+1)} := \alpha^{(r+\frac{1}{2})} \text{diag} \left(\frac{\nu}{(\alpha^{(r+\frac{1}{2})})^\top \mathbf{1}} \right)$$

	ν_1	$\dots$	ν_j	$\dots$	ν_m
μ_1	$\alpha_{1,1}$	$\dots$	$\alpha_{1,j}$	$\dots$	$\alpha_{1,m}$
$\vdots$			$\vdots$		$\vdots$
μ_i	$\alpha_{i,1}$	$\dots$	$\alpha_{i,j}$	$\dots$	$\alpha_{i,m}$
$\vdots$			$\vdots$		$\vdots$
μ_n	$\alpha_{n,1}$	$\dots$	$\alpha_{n,j}$	$\dots$	$\alpha_{n,m}$

◆ See also block-iterative SMART, mirror descent algorithm

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Sinkhorn Algorithm (Dual)

$$(D) \quad \max_{\varphi, \psi} F(\varphi, \psi) = \max_{\varphi, \psi} \langle K, e^{-P_1^\top \varphi - P_2^\top \psi} \rangle + \langle \mu, \varphi \rangle + \langle \nu, \psi \rangle$$

$$P_1 = (I_m \quad \dots \quad I_m) \in \mathbb{R}^{m \times nm}$$

$$P_2 = \begin{pmatrix} \underbrace{1 \dots 1}_m & 0 & \dots & 0 \\ 0 & \underbrace{1 \dots 1}_m & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \underbrace{1 \dots 1}_m \end{pmatrix} \in \mathbb{R}^{n \times nm},$$

Idea:

$$\nabla_{\varphi} F = -\text{diag}(e^{-\varphi}) K \text{diag}(e^{-\psi}) \mathbf{1} + \mu = 0$$

Sinkhorn algorithm with $u := e^{-\varphi}$, $v := e^{-\psi}$

$$\nabla_{\varphi} F = 0 : \text{diag}(e^{-\varphi}) K \text{diag}(e^{-\psi^{(r-1)}}) \mathbf{1} = \mu \quad \rightarrow \quad u^{(r)} = \frac{\mu}{K v^{(r-1)}}$$

$$\nabla_{\psi} F = 0 : (\text{diag}(e^{-\varphi^{(r)}}) K \text{diag}(e^{-\psi}))^\top \mathbf{1} = \nu \quad \rightarrow \quad v^{(r)} = \frac{\nu}{K^\top u^{(r)}}$$

Relation to α : (set gradient of Lagrangian wrt α to 0)

$$0 = \log \alpha - \log K - P_1^\top \varphi - P_2^\top \psi \quad \Longleftrightarrow \quad \log \alpha = \log K + P_1^\top \varphi + P_2^\top \psi$$

$$\alpha = \text{diag}(u) K \text{diag}(v)$$

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On $\mathbb{R}_{>0}^d$ consider the equivalence relation

$$x \sim x' \iff \exists t > 0 : x = t x'.$$

The quotient space, called **positive projective space**

$$\mathbb{P}_{>0}^d := \mathbb{R}_{>0}^d / \sim$$

can be endowed with a vector space structure using the following operations:

- ◆ Addition (internal operation): $x + x := x \odot y$,
- ◆ Scalar multiplication (external operation): $\gamma \cdot x := x^\gamma$,

with component-wise product $\odot$. Using the **Hilbert projective metric**:

$$d_H(x, x) := \log \left(\frac{\max_i x_i / y_i}{\min_i x_i / y_i} \right), \quad x, x \in \mathbb{P}_{>0}^d.$$

the space $\mathbb{P}_{>0}^d$ becomes a complete metric space.

Linear convergence of Sinkhorn algorithm:

$$d_H(u^{(r+1)}, u^*) \leq \lambda(K)^2 d_H(u^{(r)}, u^*)$$

with

$$\lambda(K) := \sup \left\{ \frac{d_H(Kx, Ky)}{d_H(x, y)} : x \not\sim y \right\}$$

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Further Topics

- ◆ Unbalanced optimal transport
- ◆ Sliced Wasserstein distances
- ◆ Gromov-Wasserstein distances
- ◆ Partial Optimal Transport
- ◆ ...

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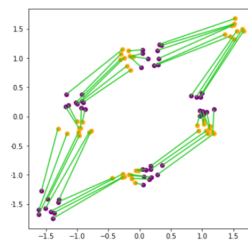
Given:

- ◆ positive measures $\mu, \nu \in \mathcal{M}_+(\mathbb{R}^d)$.
- ◆ cost function $c : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}_+$.
- ◆ divergence measure $D_f(\cdot|\cdot)$ (e.g., KL-divergence, Total Variation).
- ◆ regularization parameters $\tau_1, \tau_2 > 0$.

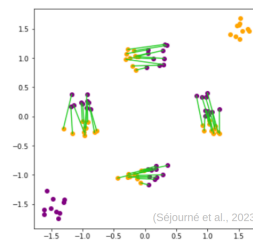
Find **optimal coupling** α :

$$\begin{aligned} \text{UOT}(\mu, \nu) = \min_{\alpha \in \mathcal{M}_+(\mathbb{R}^d \times \mathbb{R}^d)} & \int_{\mathbb{R}^d \times \mathbb{R}^d} c(x, y) d\alpha(x, y) \\ & + \tau_1 D_f((P_1)_\# \alpha | \mu) + \tau_2 D_f((P_2)_\# \alpha | \nu) \end{aligned}$$

- ◆ Compares measures with different mass.
- ◆ Robust to outliers.



(a) OT matching



(b) Unbalanced OT matching

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Given: $\mu, \nu \in \mathcal{P}_p(\mathbb{R})$

Define 1d Projection: $P_\theta(x) = \theta^\top x$ for $\theta \in \mathbb{S}^{d-1}$ (unit sphere)
 $\Rightarrow P_{\theta, \#}\mu, P_{\theta, \#}\nu \in \mathcal{P}_p(\mathbb{R})$

Introduce:

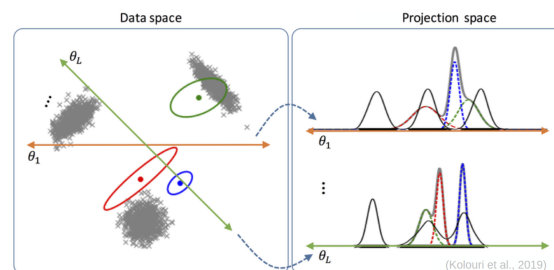
$$SW_p^p(\mu, \nu) = \int_{\mathbb{S}^{d-1}} W_p^p(P_{\theta, \#}\mu, P_{\theta, \#}\nu) d\xi(\theta),$$

where $\xi \sim \mathcal{U}(\mathbb{S}^{d-1})$ is the uniform measure on the unit sphere $\mathbb{S}^{d-1}$.

Calculate Monte Carlo Estimate:

$$SW_p^p(\mu, \nu) \approx \sum_{l=1}^L W_p^p(P_{\theta_l, \#}\mu, P_{\theta_l, \#}\nu), \quad \theta_l \sim \mathcal{U}(\mathbb{S}^{d-1}).$$

- ◆ Fast 1d Wasserstein Computation via Sorting ($\mathcal{O}(n \log n)$)
- ◆ Fast SW approximation ($\mathcal{O}(L n \log n)$)



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Given metric measure spaces:

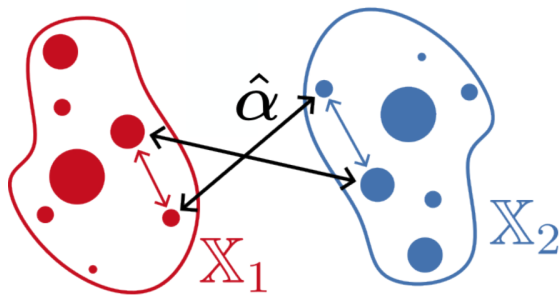
$$\mathbb{X} = (X, d_X, \mu), \quad (X, d_X) \text{ compact metric space, } \mu \in \mathcal{P}(X),$$

$$\mathbb{Y} = (Y, d_Y, \nu), \quad (Y, d_Y) \text{ compact metric space, } \nu \in \mathcal{P}(Y).$$

Find optimal **GW transport plan** $\hat{\alpha} \in \mathcal{P}(X \times Y)$:

$$\hat{\alpha} \in \operatorname{argmin}_{\alpha \in \Pi(\mu, \nu)} \iint_{(X \times Y)^2} |d_X(x, x') - d_Y(y, y')|^2 d\alpha(x, y) d\alpha(x', y').$$

$$\operatorname{GW}^2(\mathbb{X}, \mathbb{Y}) = \iint_{(X \times Y)^2} |d_X(x, x') - d_Y(y, y')|^2 d\hat{\alpha}(x, y) d\hat{\alpha}(x', y').$$



→ GW vanishes only for isomorphic spaces and allows for interpolating geometries.

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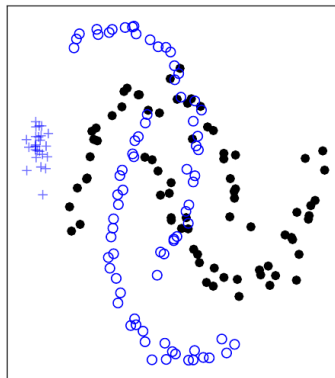
For positive μ, ν , and $s \leq \min\{\|\mu\|_{\text{TV}}, \|\nu\|_{\text{TV}}\}$, consider the **partial plans**

$$\Pi_{\leq}(\mu, \nu) := \left\{ \alpha \in \mathcal{M}_+(X \times Y) : (P_1)_{\#}\alpha \leq \mu, (P_2)_{\#}\alpha \leq \nu, \|\alpha\|_{\text{TV}} = s \right\}.$$

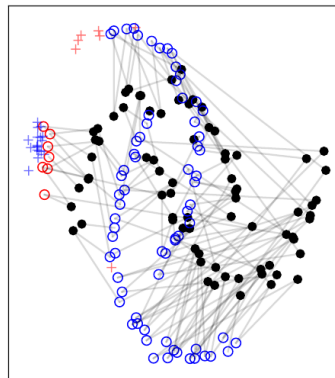
Partial (Gromov–)Wasserstein Distance:

$$\text{PW}_p^p(\mu, \nu) := \inf_{\alpha \in \Pi_{\leq}(\mu, \nu)} \int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|^p d\alpha(x, y)$$

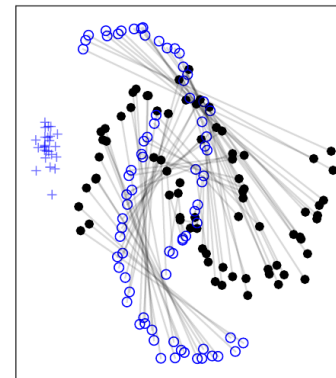
$$\text{PGW}_p^p(\mathbb{X}, \mathbb{Y}) := \inf_{\alpha \in \Pi_{\leq}(\mu, \nu)} \iint_{(X \times Y)^2} |d_X(x, x') - d_Y(y, y')|^p d\alpha(x, y) d\alpha(x', y')$$



source/target



Wasserstein

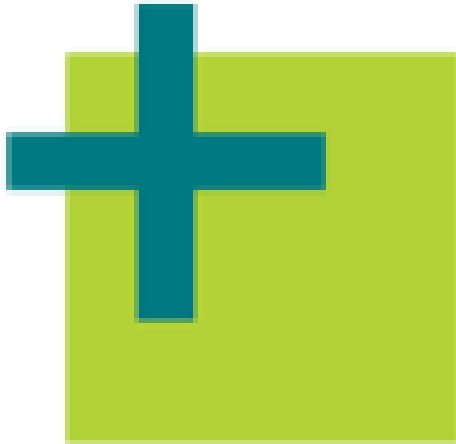


Gromov–Wasserstein

(Chapel et al., 2020)

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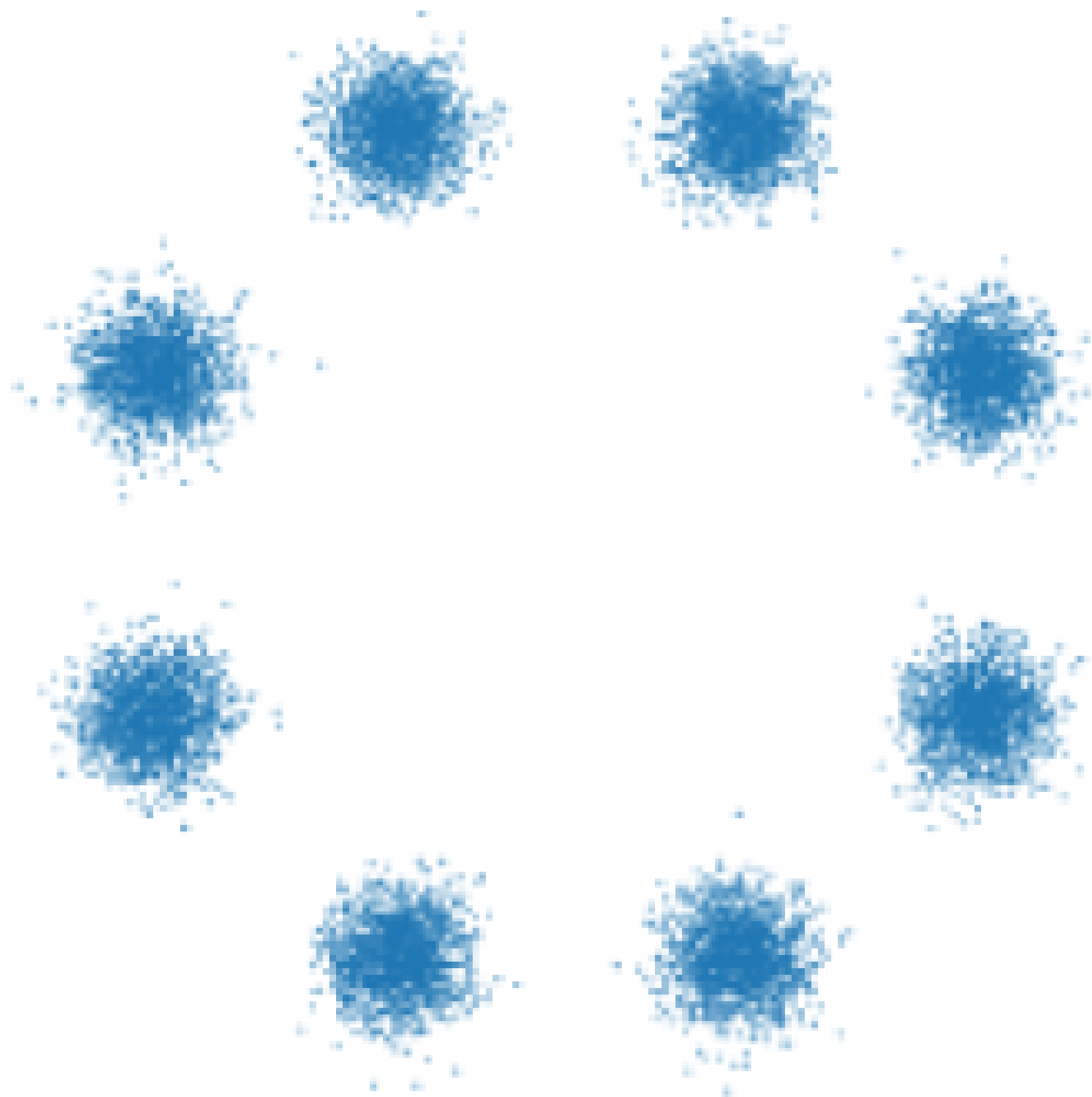
•
$$\sum_{j=1}^7 \mu_j \delta_{x_j} = \mu$$

•
$$\sum_{k=1}^8 \nu_k \delta_{y_k} = \nu$$

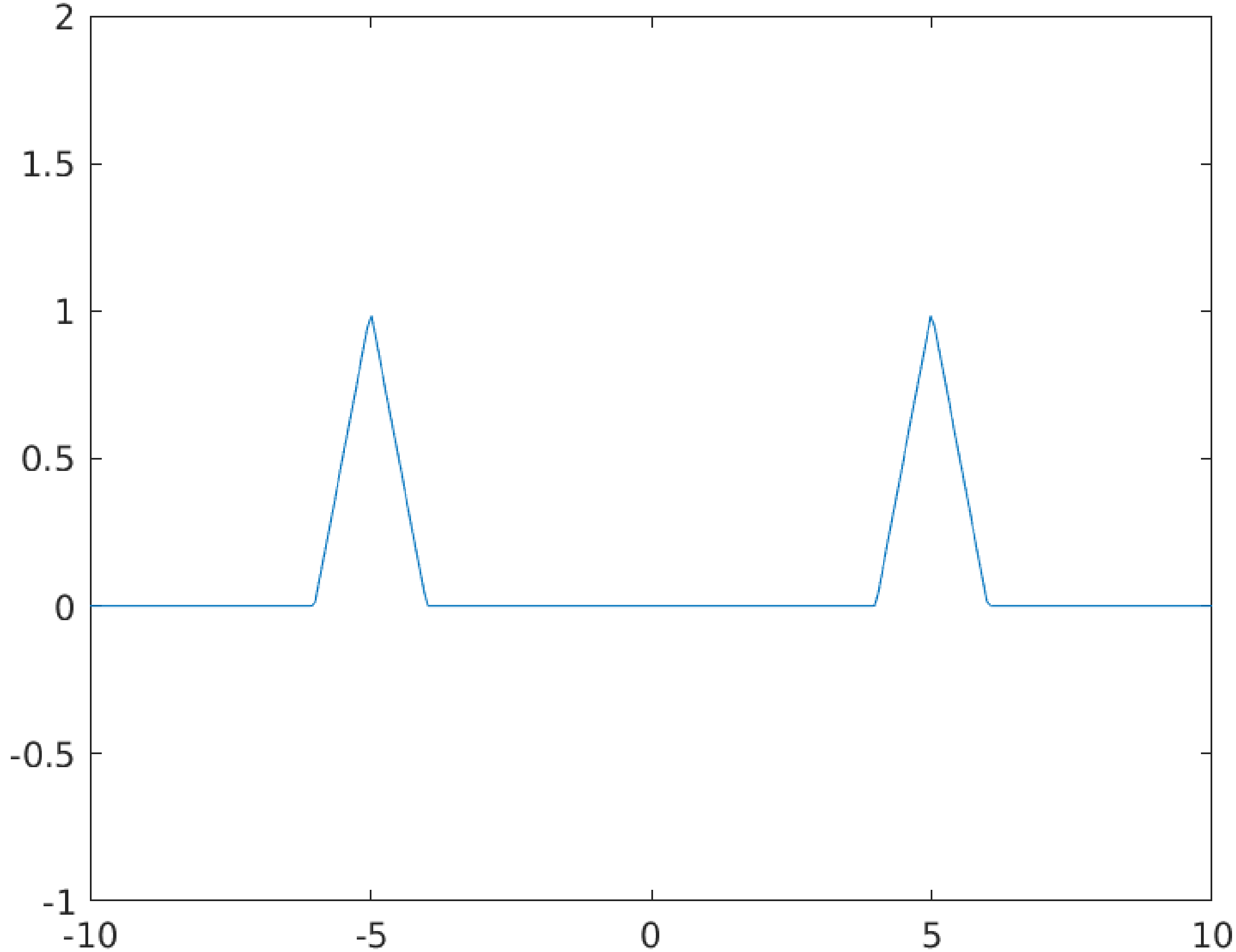


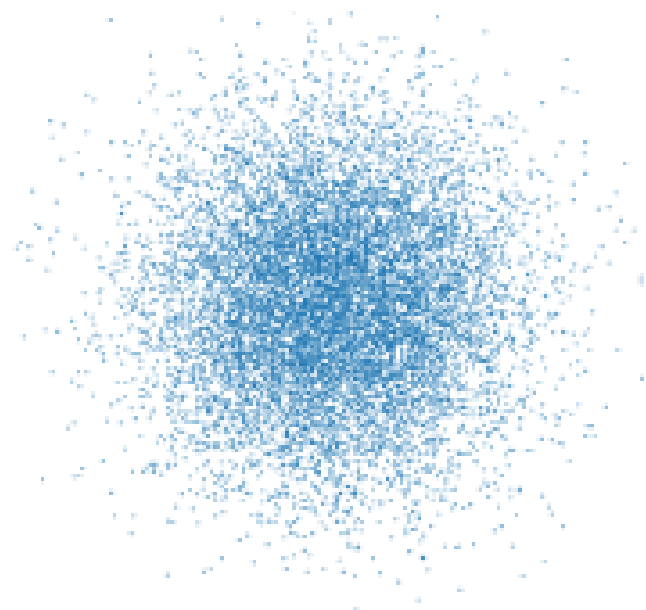
•
$$\sum_{j=1}^{17} \mu_j \delta_{x_j} = \mu$$

•
$$\sum_{k=1}^5 \xi_k \delta_{z_k} = \xi$$

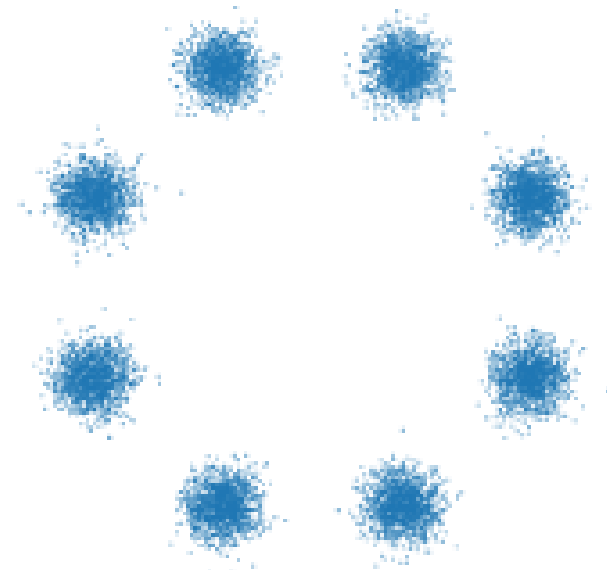
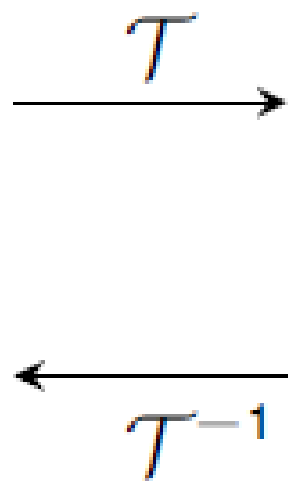


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8	9	1	2	5	0	0	6	6	4
6	7	0	1	6	3	6	3	7	0
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7	9	6	4	7	0	6	9	2	3



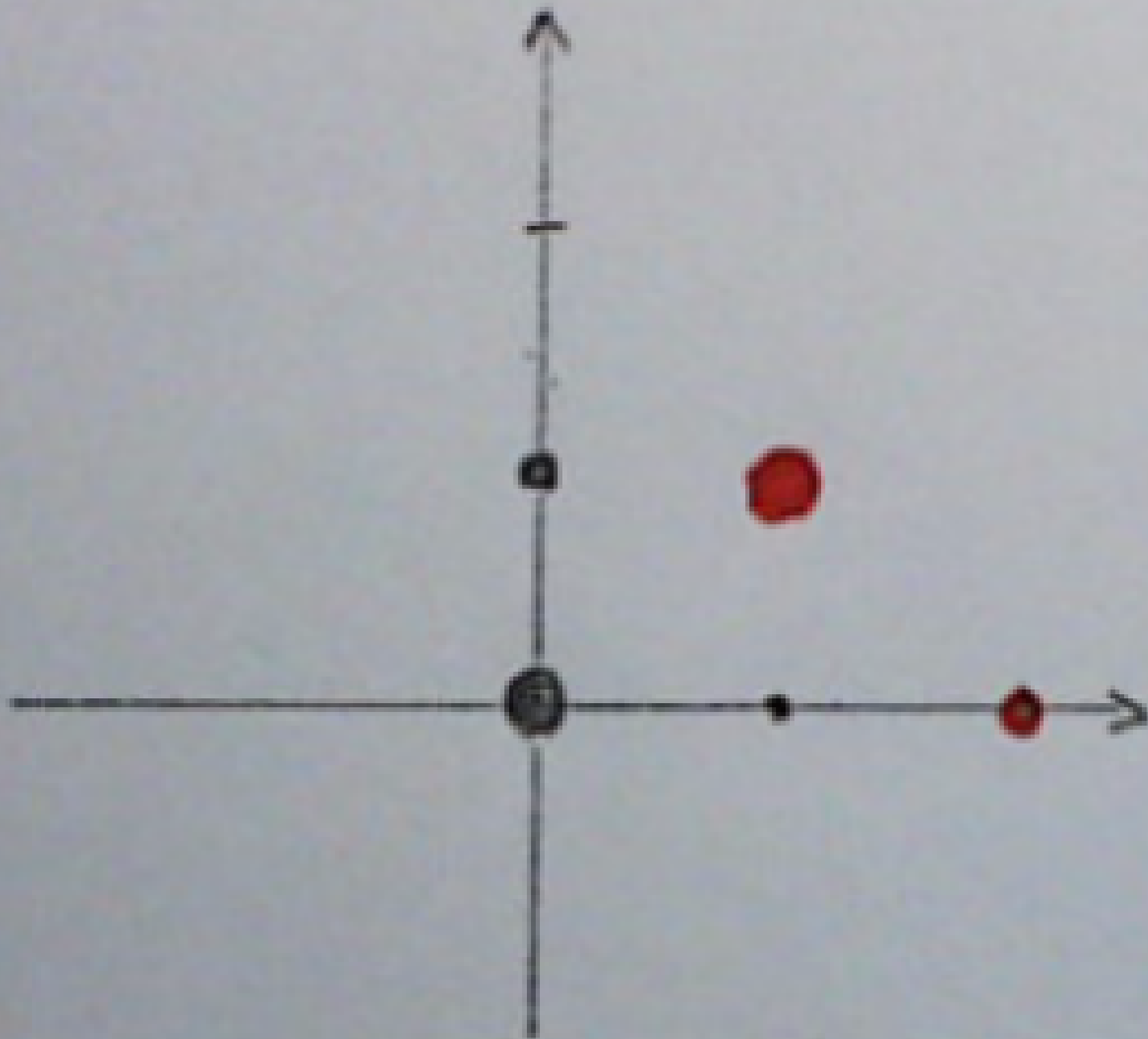


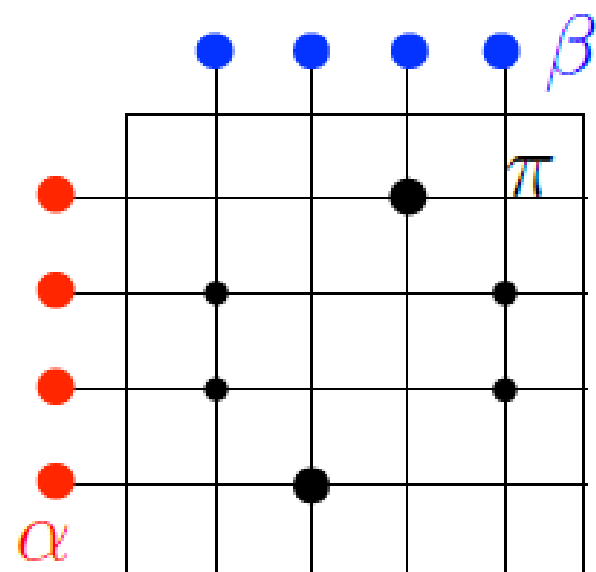
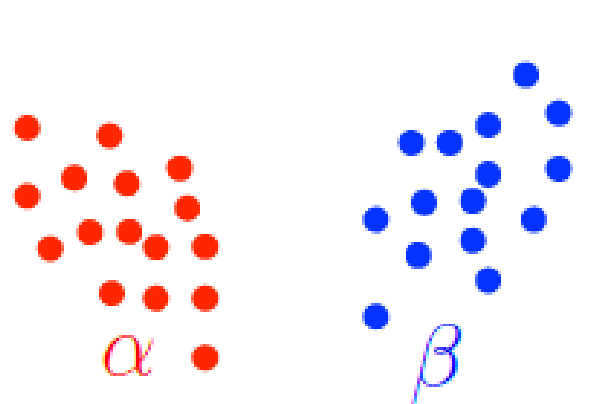
Latent distribution P_Z



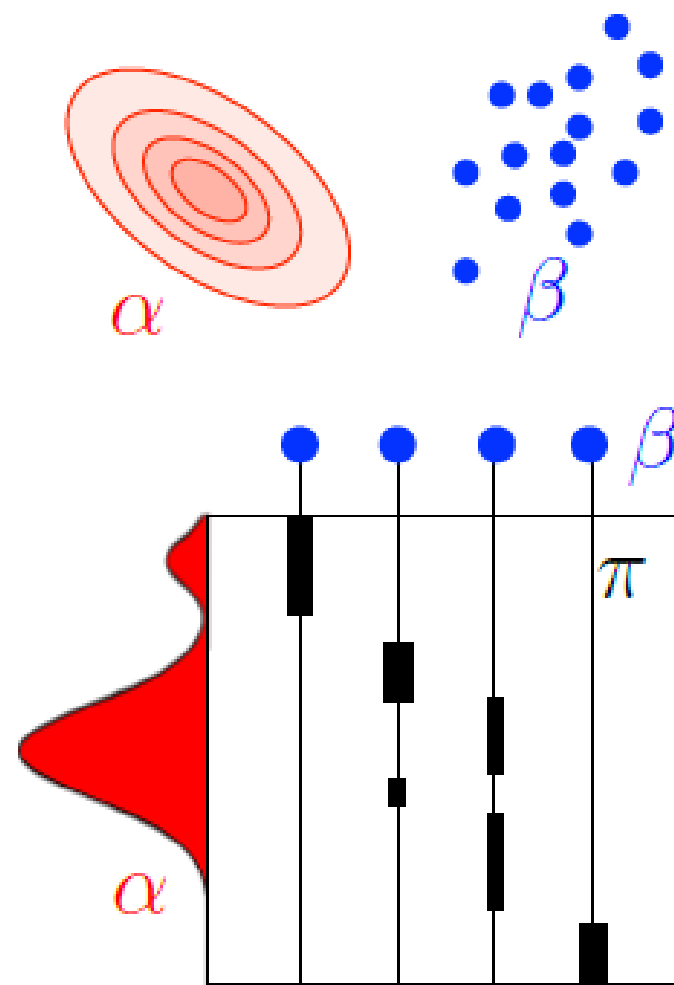
Data distribution P_X



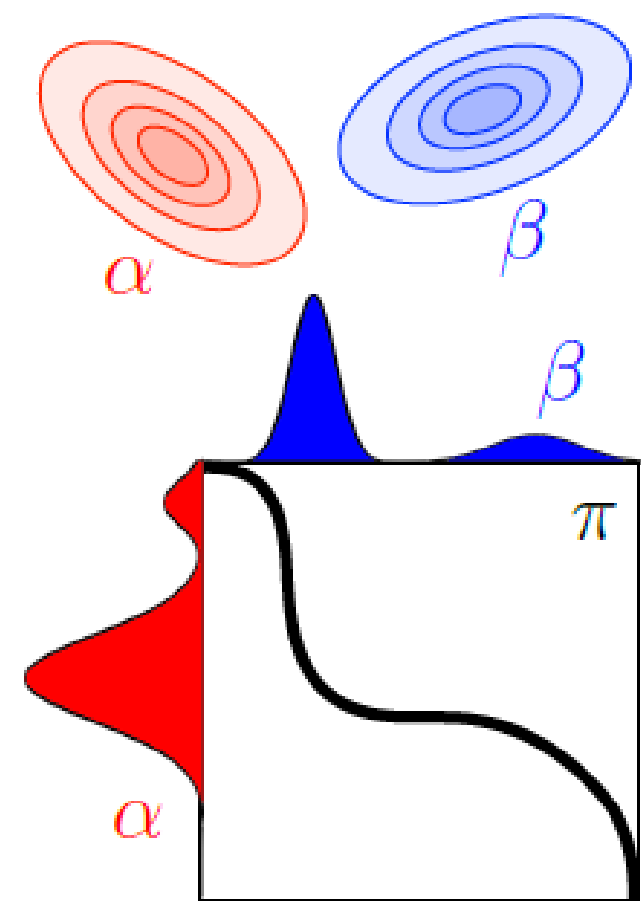




Discrete

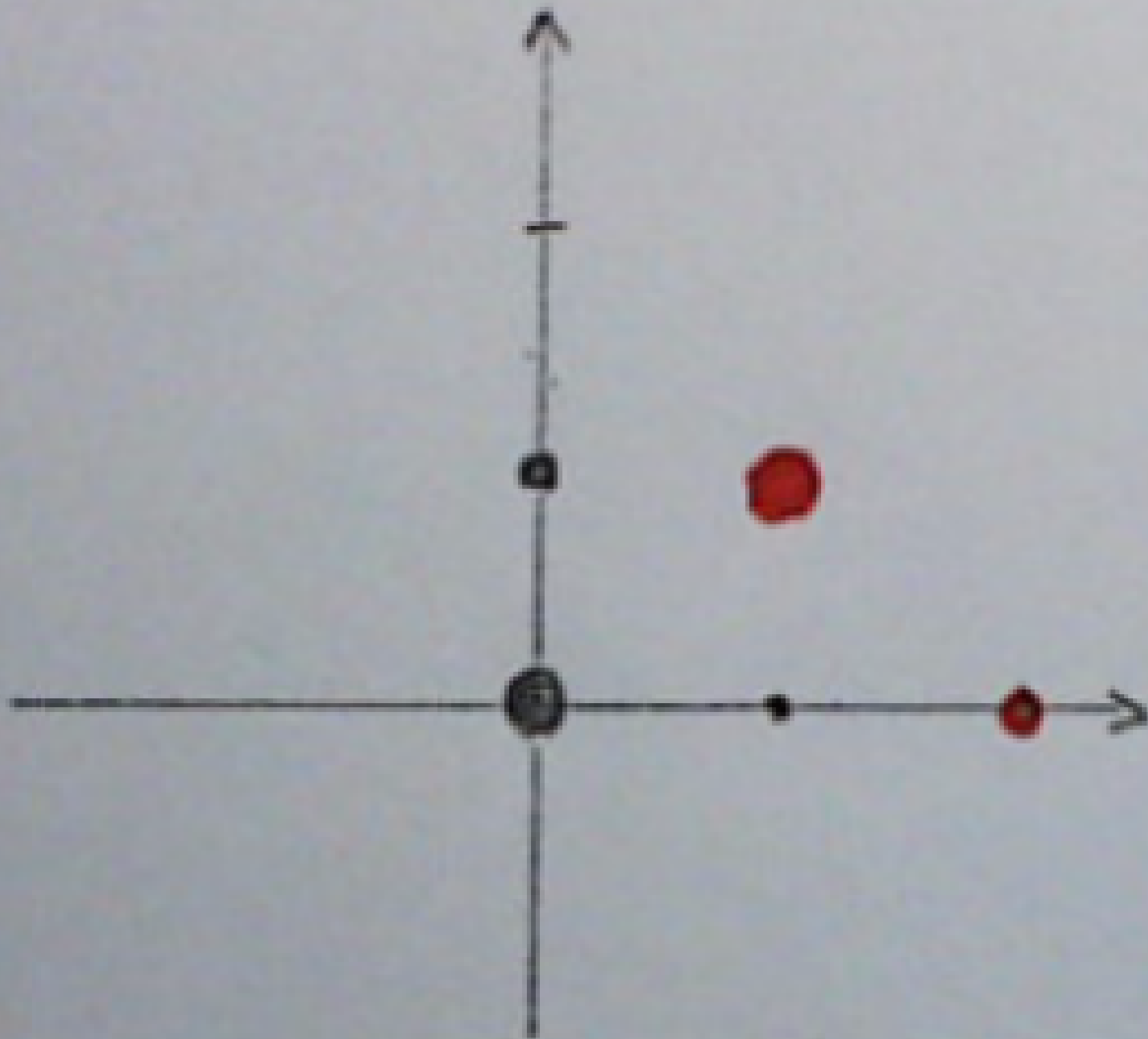


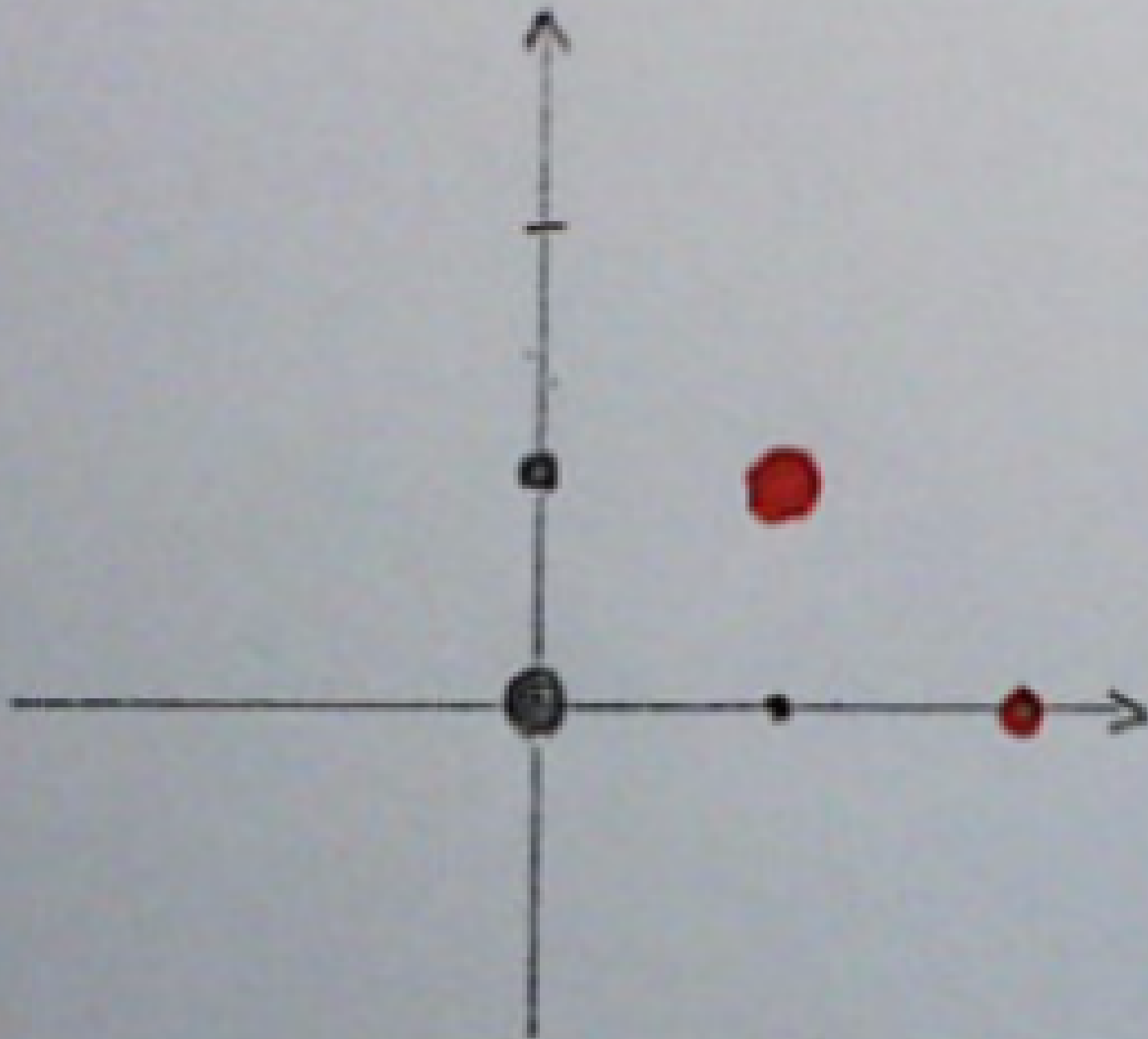
Semidiscrete

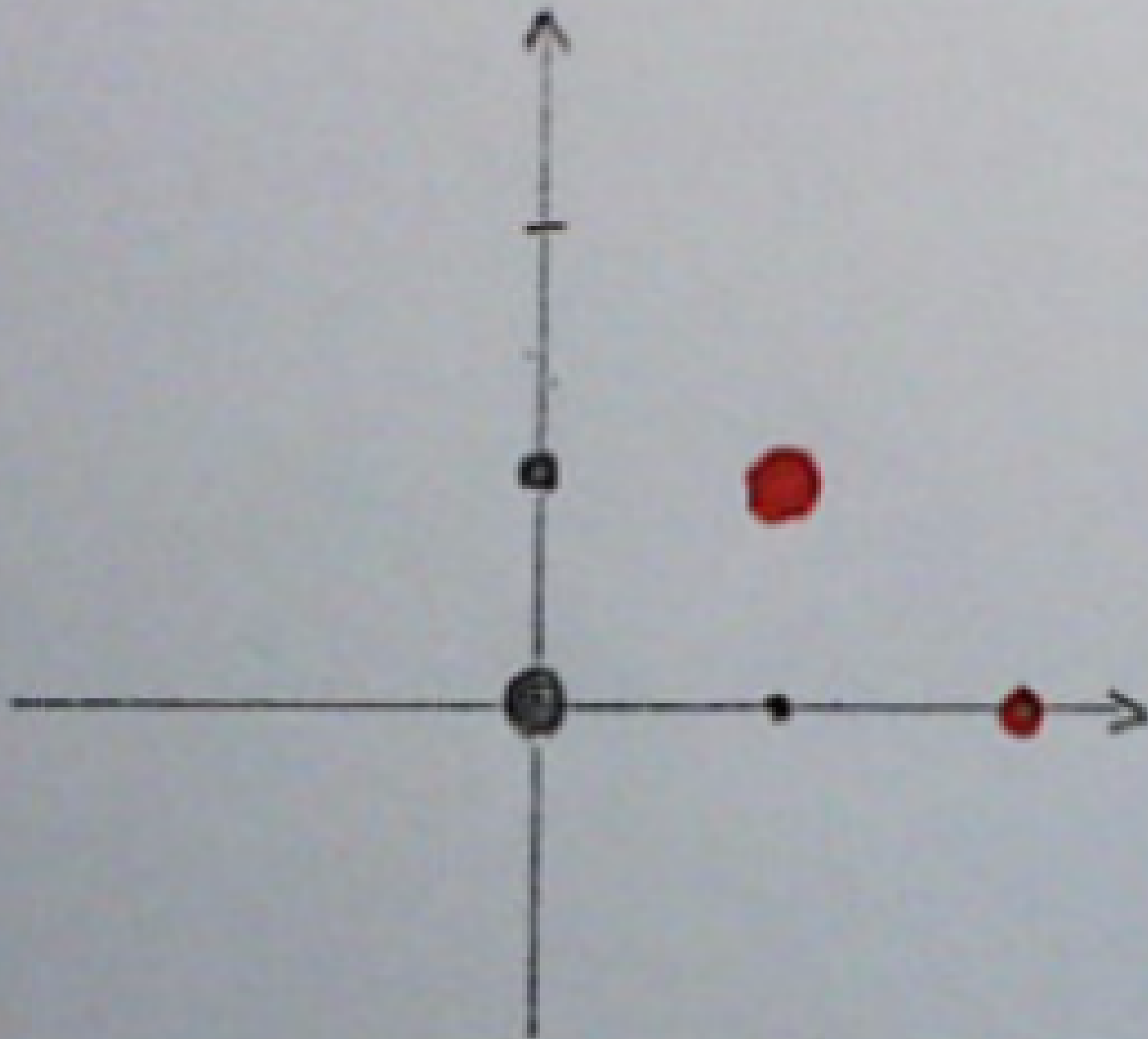


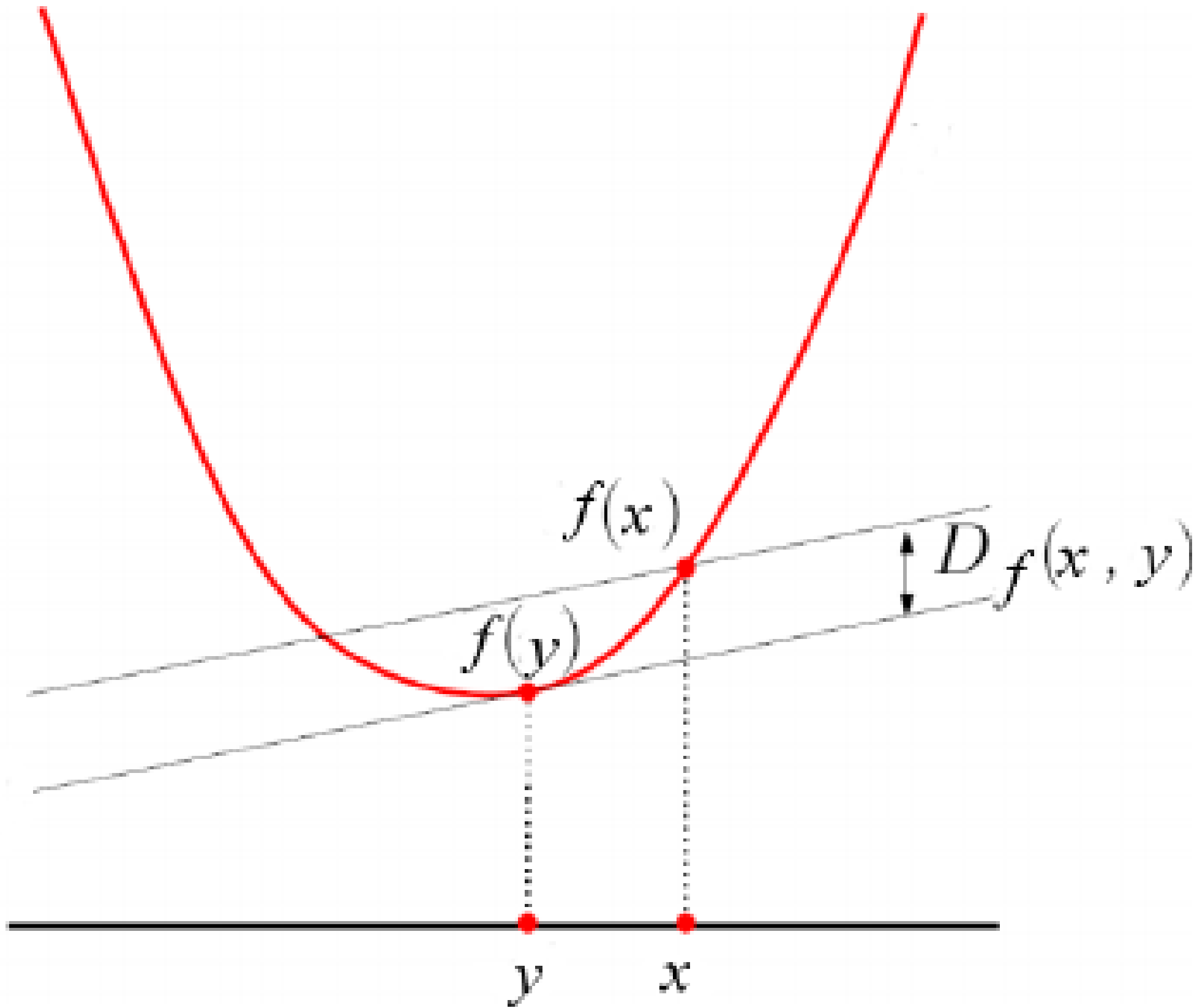
Continuous



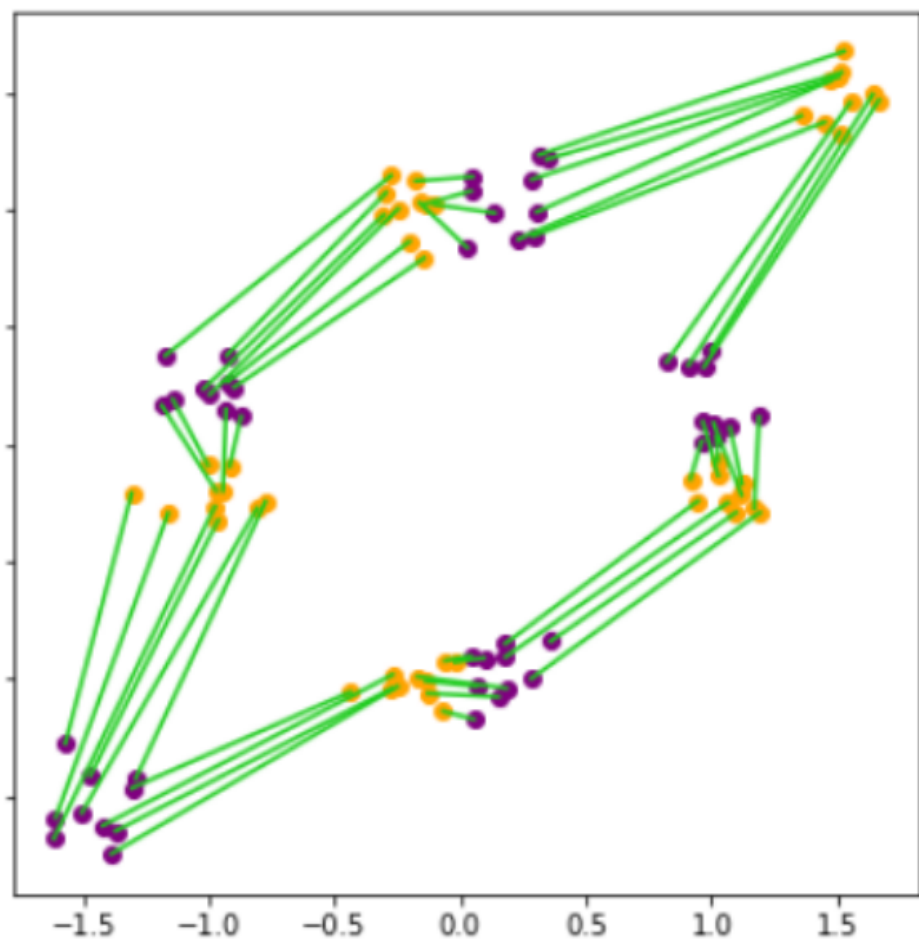




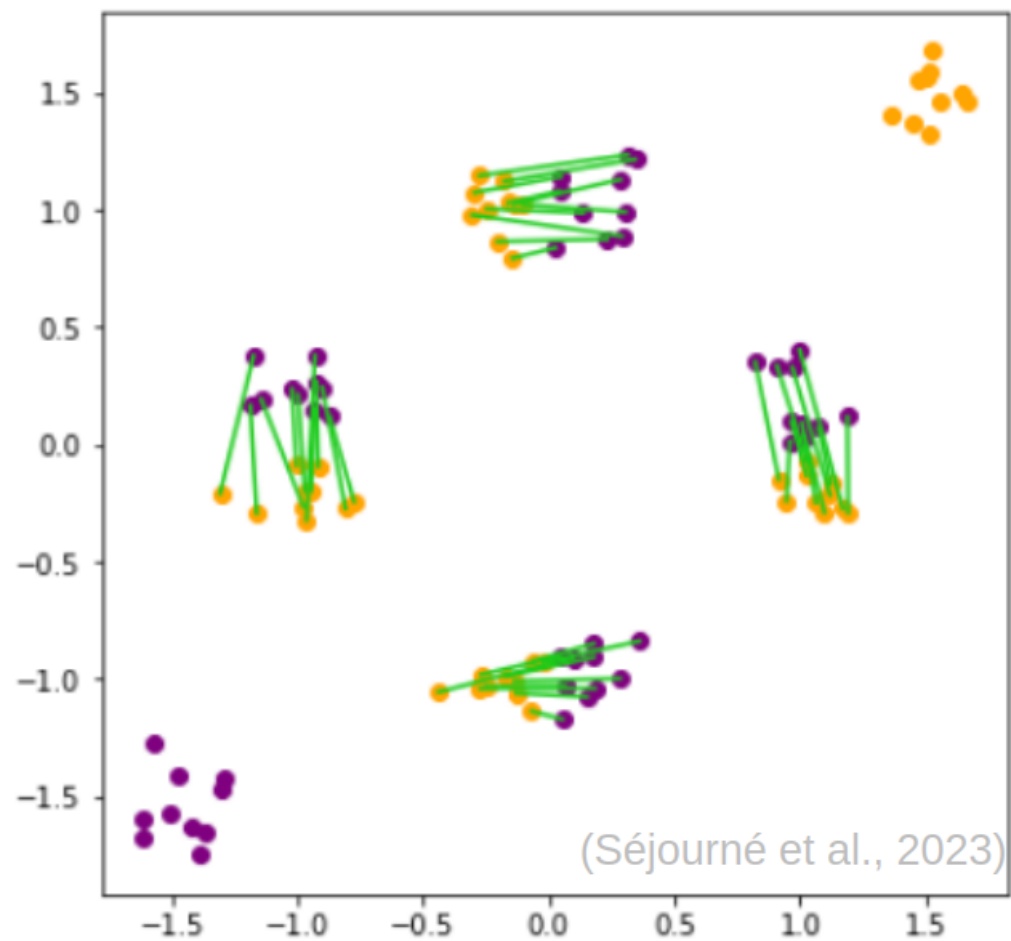




Function Name	$f(x)$	$\text{dom } f$	$D_f(x, y)$
Squared Norm	$\frac{1}{2}x^2$	$(-\infty, \infty)$	$\frac{1}{2}(x - y)^2$
Shannon Entropy	$x \log x$	$[0, \infty)$	$x \log \frac{x}{y} - x + y$
Bit Entropy	$x \log x + (1 - x) \log(1 - x)$	$[0, 1]$	$x \log \frac{x}{y} + (1 - x) \log \frac{1-x}{1-y}$
Burg Entropy	$-\log x$	$(0, \infty)$	$\frac{x}{y} - \log \frac{x}{y} - 1$



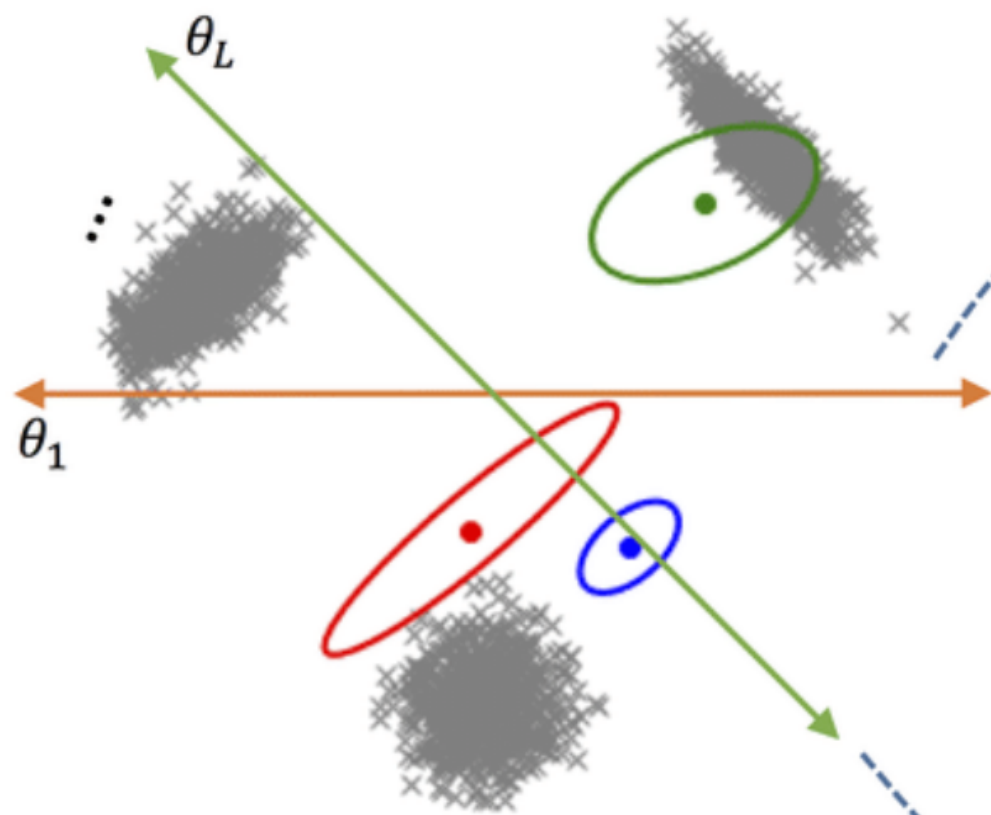
(a) OT matching



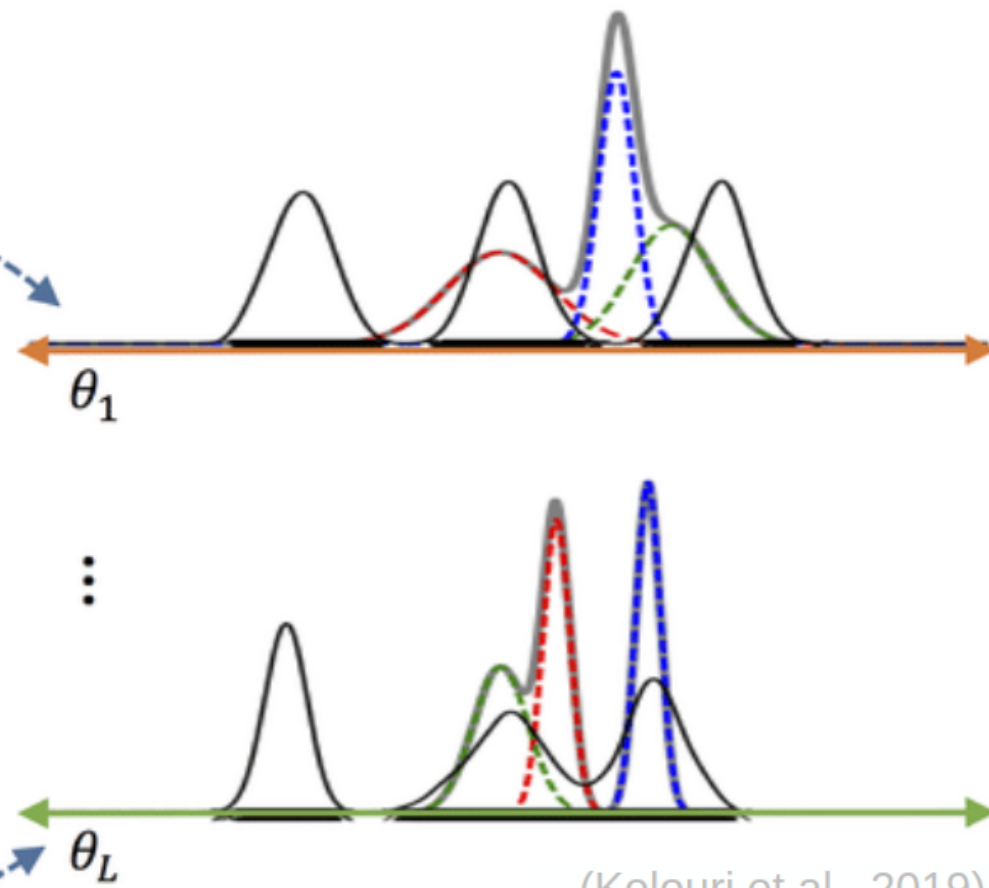
(b) Unbalanced OT matching

(Séjourné et al., 2023)

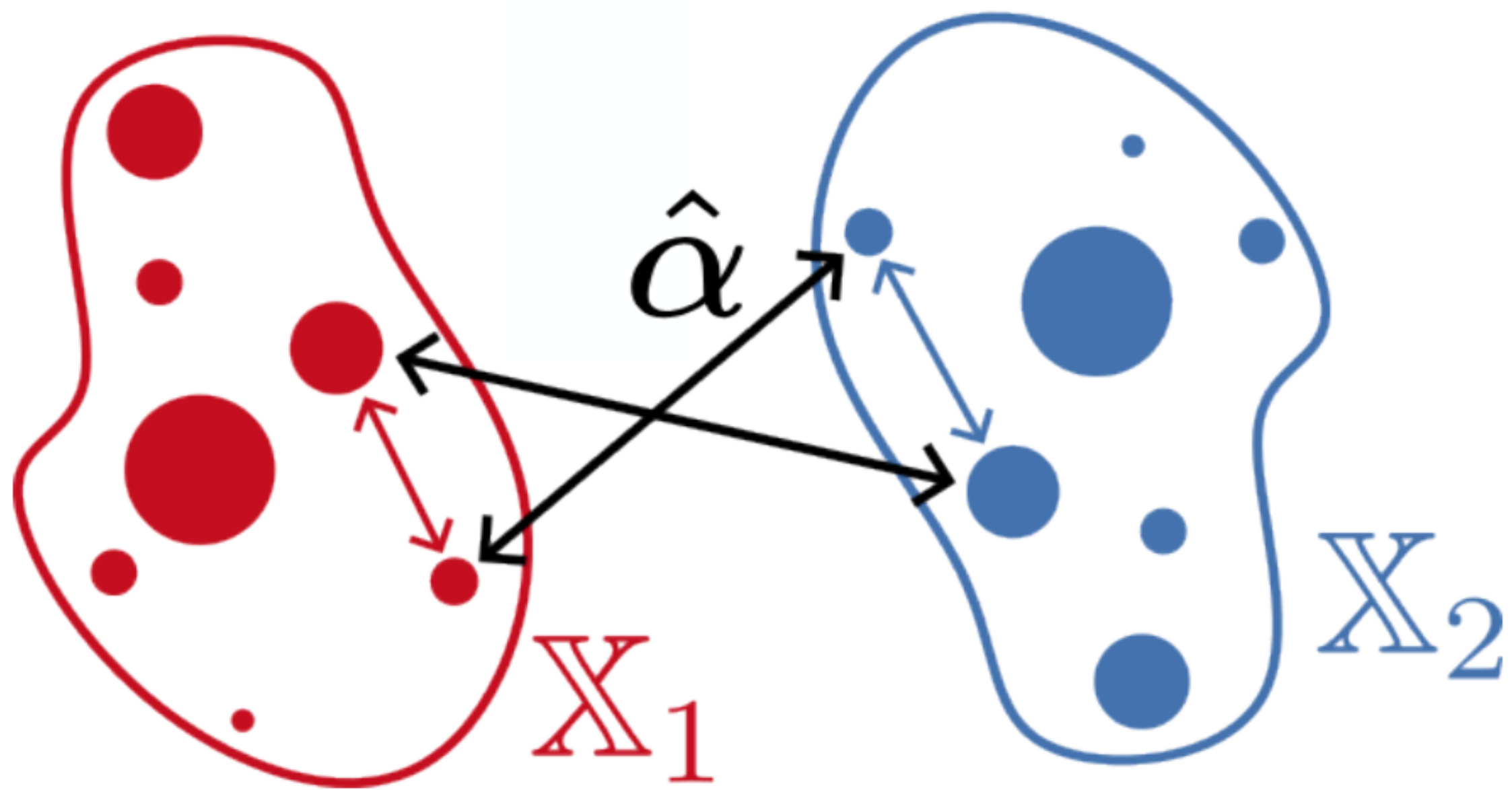
Data space



Projection space



(Kolouri et al., 2019)



1000 x 400



(Beier & Beinet, 2025)

